بسم الله الرحمن الرحيم

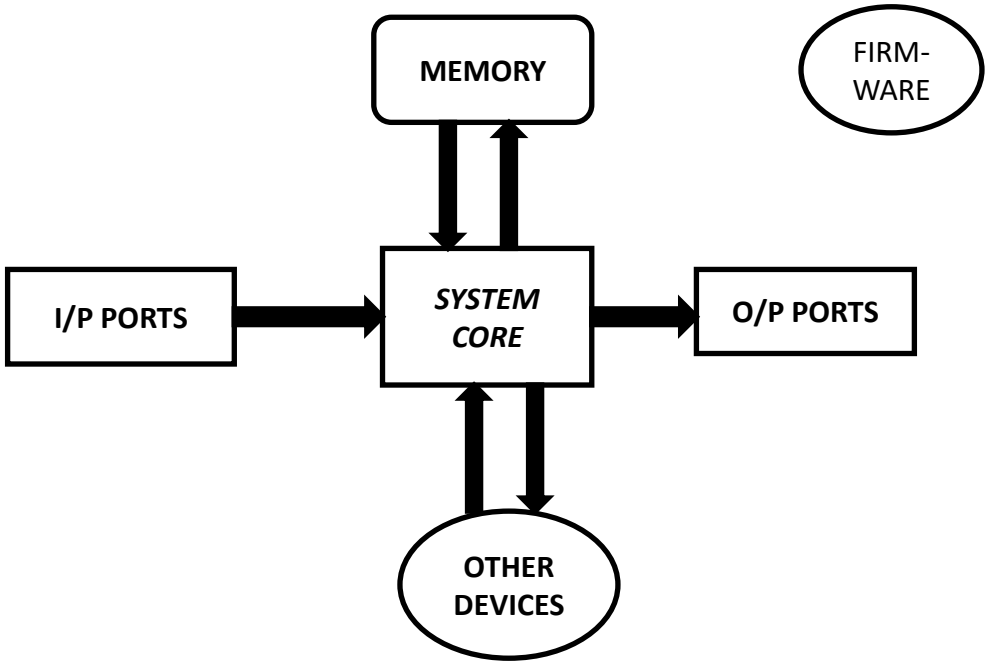
Embedded Systems - COE349

Dr. Khalid Ammar
Office: 154-4 .
Phone: 705-6616

Embedded System

An embedded system special-purpose computer system, it could be stand alone or a part of a larger system.

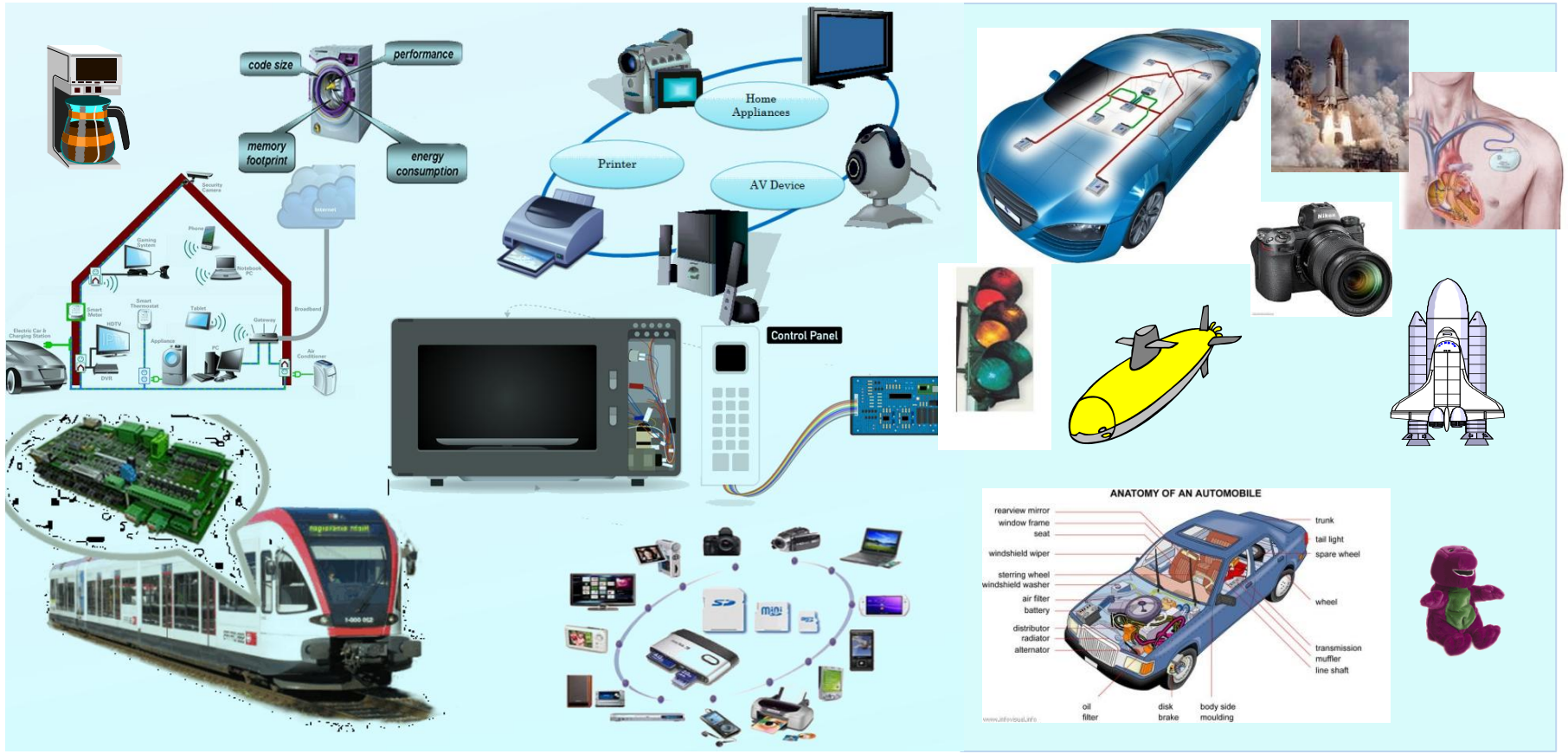
- Embedded systems
  - The Core of embedded systems :
    - ✓ General purpose and domain specific processor
      - Microprocessors
      - Microcontrollers.
      - Digital Signal Processors
    - ✓ Application Specific Integrated Circuits. (ASIC)
    - ✓ Programmable logic devices(PLD's)
  - Memory
  - Sensors & Actuators
  - Communication Interface
  - Embedded Firmware (software that provides the low-level control for a device's specific hardware.)
  - Other System Components



- Almost 80% of the embedded systems are processor/ controller based.
- The processor may be microprocessor or a microcontroller or DSP processor, depending on the domain and application.
- The majority of the embedded product uses a microcontroller to do one task only.

An embedded system special-purpose computer system, it could be stand alone or a part of a larger system.

- ✓ Embedded system are more than humans on the planet, already 40 billion of such devices by 2020
- ✓ 99% of processors used in embedded systems
- ✓ €The Market size is about >100 times the desktop market



Course Catalog Description:

This course introduces the hardware and software design of embedded systems using microcontrollers. Students are introduced to microcontroller programming in both assembly and C. Important subsystems of the microcontroller are covered such as timers, interrupts, serial transmission of data, analog to digital and digital to analog converters. There are a series of exercises introduced into the lectures and labs which give students hands-on experience with working with microcontroller. At the end of the course, each student will choose a design project to work on during the last few weeks.

Course Learning Outcomes:

After completing this course, students will be able to:

a. Design hardware interfaces for various input/output devices

b. Develop microcontroller Programs for embedded systems applications

c. Apply new knowledge related to embedded systems

d. Engage in effective communication, documentation and presentation.

e. Function effectively in a team

f. Conduct experiments using CAD tools to simulate and verify embedded systems.

Week	Topics and Contents
1	Introduction to embedded systems. Microcontroller Architecture, Memory Organization,
2	Instruction Set. IO Ports Interfacing and Programming
3	Programming microcontrollers in Assembly
4	Programming microcontrollers in C language
5	Microcontroller PIN connections, Microcontroller clock and power connections
6	Programming and Interfacing Microcontroller with LEDs, pushbuttons, 7 segment displays and LCD. Custom Character generation.
7	Programming and Interfacing Microcontroller with switches and Keypads
8	PIC ADC architecture, Sensors & Actuators,
9	PIC ADC applications.
10	Programming and Interfacing Microcontroller with (DC motors, stepper motors and Servo Motors). Relays as a driver.
11	PIC timers/counters architecture and applications
12	PIC Capture/Compare/PWM and programming
13	PIC Serial Communication, EUSART and applications
14	Data acquisition systems and Signal Conditioning
15	Course project presentation and discussion

Course materials

Textbook:

Title: PIC Microcontroller and Embedded Systems: Using Assembly and C for Pic18

Authors: Muhammad Ali Mazidi, Danny Causey, Rolin McKinlay

Publisher: MicroDigitalEd, 2016

ISBN: 099792599X, 9780997925999

Title: A Complete Notebook on PIC Microcontrollers

Authors: Milan VerlePublisher: MicroDigitalEd, 2016

ISBN: 1794803440, 9781794803442

Rules/Regulations

- Arrive on time

- Attendance in all lectures is mandatory.
  - If the student misses more than (10% of class time), a warning is issued.
  - More than (20 % of class time), a warning from the dean will be issued.
  - More than (25% of class time) will result in an automatic F in the course.

- Mobile phones must be turned off before during the class room/laboratory.

- Don't make noise, no side talk

Grading

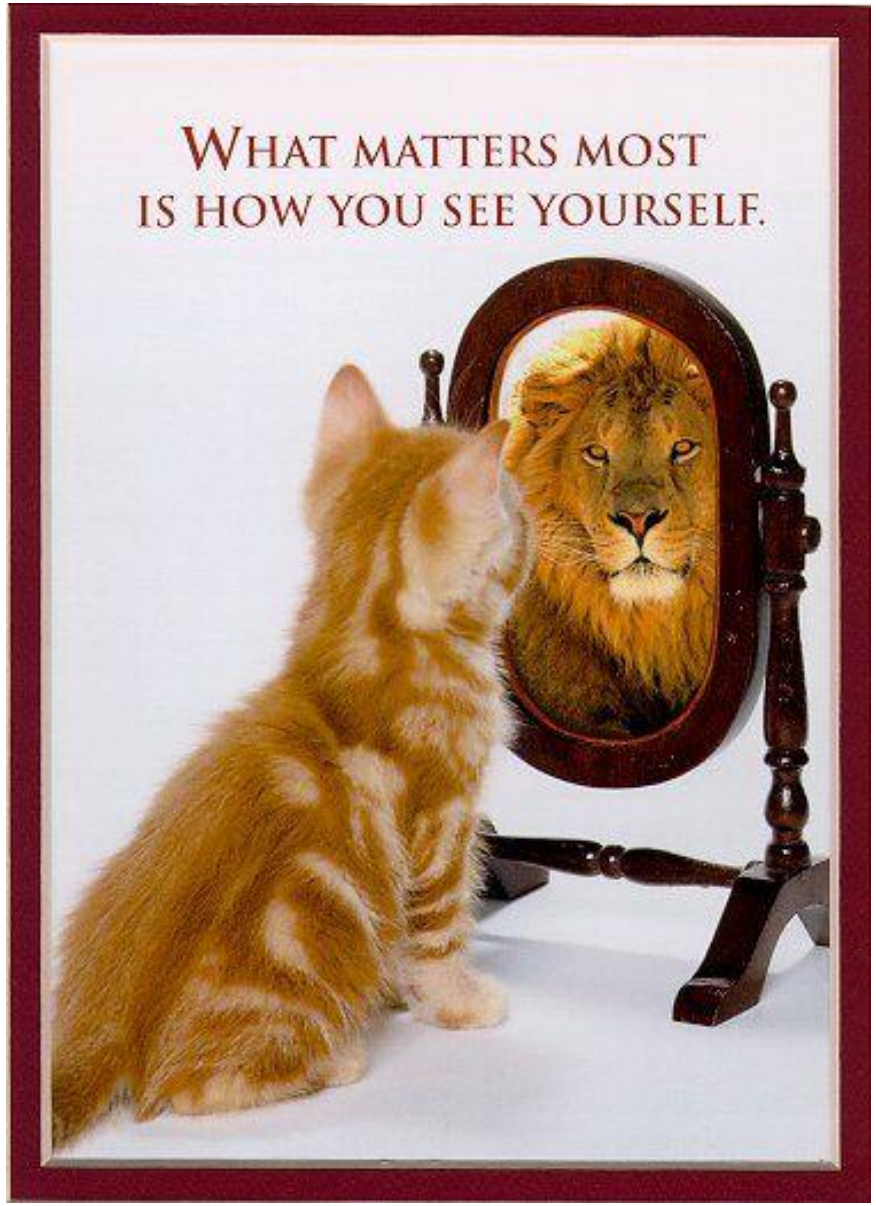
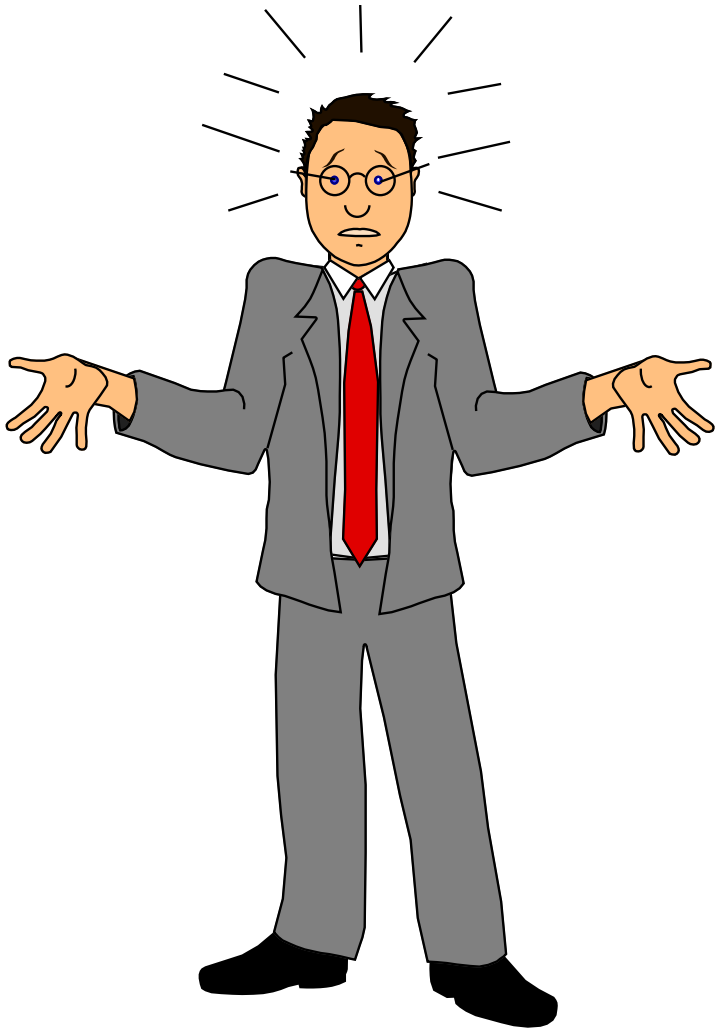
• Tests	15%
• Project	10 %
• Mid-term examination	20 %
• Coursera certificate	5%
• Lab activities	25 %
• Final examination	30 %

Pass level for all examinations is 60% !

Few Recommendations

- If you are not sure: ask!
- Talk to me after class, come to office hours
- Early communication solves problems easiest
- Don't wait until it's too late

- See yourself for who you can become, not just who you are today.
- Confidence
- Self-Belief



Questions